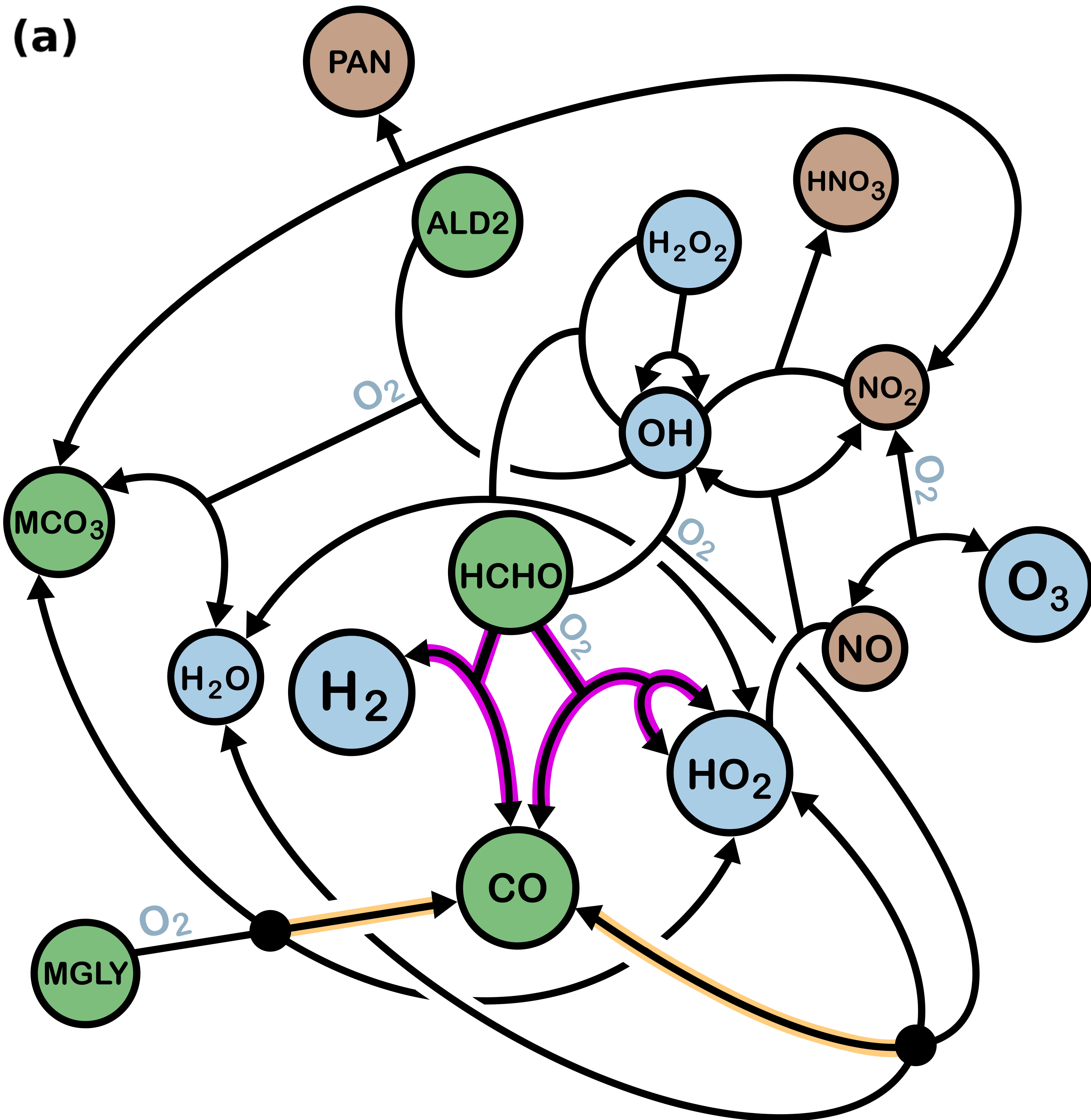




$$L_c = [\text{HCHO}] + [\text{CO}] + 2[\text{ALD2}] + 3[\text{MGLY}] + 2[\text{MCO}_3] + 2[\text{PAN}]$$

$$L_N = [\text{NO}] + [\text{NO}_2] + [\text{HNO}_3] + [\text{PAN}]$$

$$L_H = 2[\text{HCHO}] + [\text{HO}_2] + 2[\text{H}_2\text{O}_2] + [\text{OH}] + [\text{HNO}_3] + 2[\text{H}_2] + 4[\text{ALD}_2] + 4[\text{MGLY}] + 3[\text{MCO}_3] + 3[\text{PAN}] + 2[\text{H}_2\text{O}]$$

$$L_O = 3[\text{O}_3] + [\text{NO}] + 2[\text{NO}_2] + [\text{HCHO}] + 2[\text{HO}_2] + 2[\text{H}_2\text{O}_2] + [\text{OH}] + 3[\text{HNO}_3] + [\text{CO}] + [\text{ALD2}] + 2[\text{MGLY}] + 3[\text{MCO}_3] + 5[\text{PAN}] + [\text{H}_2\text{O}] + 2[\text{O}_2]$$

$$L_{\text{subpool}} = [\text{HCHO}] + [\text{CO}] + [\text{MGLY}]$$

$$L_{\text{kinetic}} = \alpha_4[\text{O}_3] - \alpha_4[\text{NO}] + \alpha_4[\text{HO}_2] - 2\alpha_3[\text{H}_2] - \alpha_4[\text{ALD2}] + \alpha_4[\text{MGLY}] - \alpha_4[\text{H}_2\text{O}]$$

